

SNAPDEAL PROJECT

Customer Purchasing Behaviour and Product Recommendation

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- Batch :- February 2026
- Course :- Data Science
- Video Link:- [Video Link](#)

Project Overview

1. Project Overview

This project looks at how Snapdeal customers shop, what they buy, how often they browse, how satisfied they are, and how useful they find product recommendations. The data was also used to make customer groups and find areas where the shopping experience can be improved.

Project Purpose

The main purpose is to understand customer behavior and find simple ways to improve customer engagement, satisfaction and repeat purchases.

Objective

- Understand Snapdeal customer shopping behavior.
- Find useful customer groups.
- Study satisfaction, reviews and recommendations.
- Identify areas that may affect purchases and customer retention.
- Suggest practical actions based on the data.

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Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab

Project Workflow

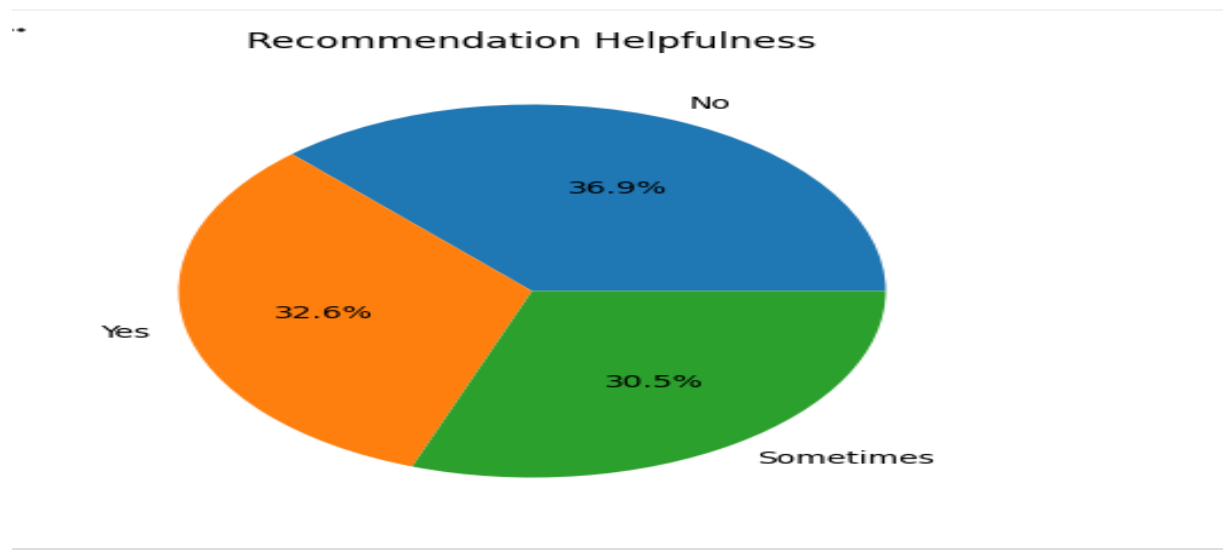
- Data cleaning
- Customer behavior analysis
- Customer segmentation
- Recommendation and review analysis
- Data visualization
- Insights and recommendations

2. Data Preparation

- 800 customer records and 24 columns were analyzed.
- No duplicate rows were found.
- 149 missing values in Product_Search_Method were filled with 'others'.
- Score-based fields such as satisfaction and rating accuracy were kept in numeric form for analysis.

Task 2 – Customer Behavior

- **Recommendation helpfulness**



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Actual output:

Mean = 1.95 , Median = 2.
No = 295, Sometimes = 244, Yes = 261.

Insight: Most customers find the recommendations only sometimes helpful. This shows that the current recommendations are not matching customer needs well.

Recommendation: Make recommendations more useful and better matched to what customers actually want.

- **Purchase Frequency:** 166 customers purchase **multiple times a week**, which is the most common purchase frequency.
- **Cart Abandonment:** **219 customers** said they leave the cart because they find a **better price elsewhere**.
- **Shopping Satisfaction:** The **mean satisfaction is 2.99** and the **median is 3**, showing that most customers are around the middle satisfaction level.

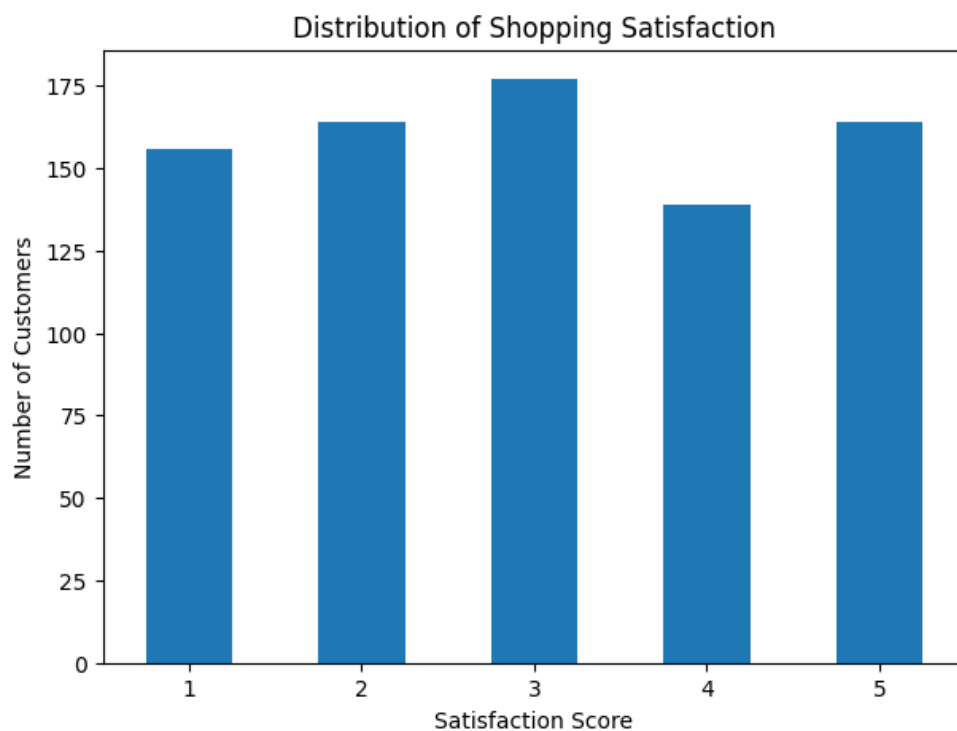
1. Shopping Satisfaction

Insight:

- Most customers are around the middle satisfaction level (**3/5**).

Recommendation:

- Improve product quality, pricing, and shopping experience to increase satisfaction.



Insight:

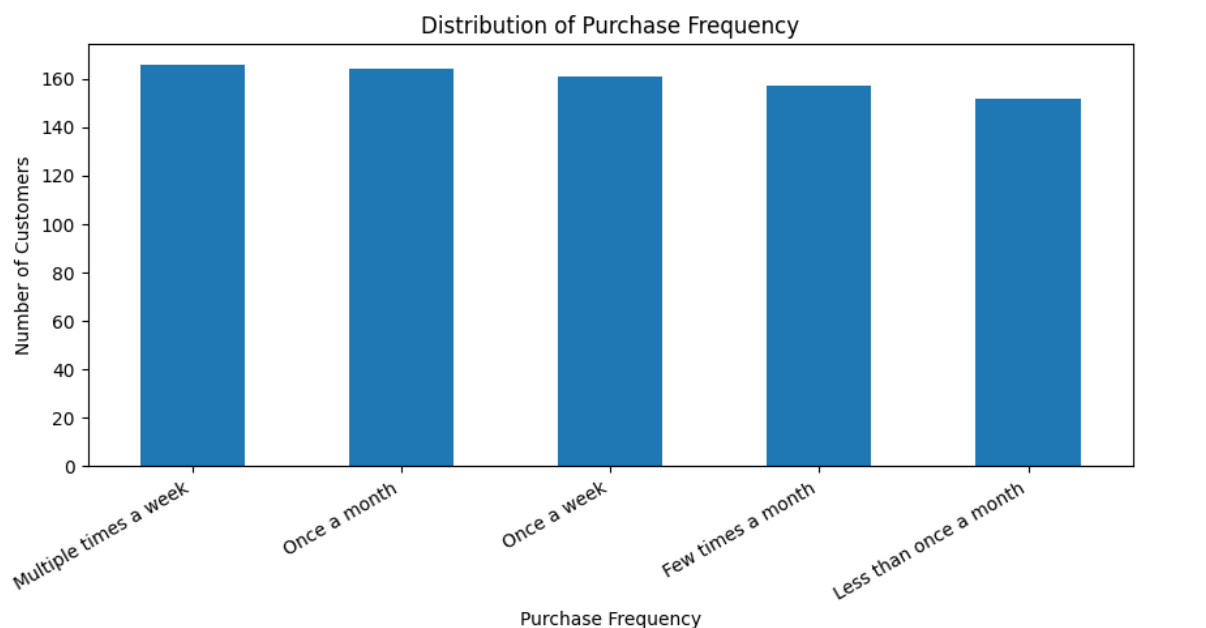
- Customers are active, but price and other practical issues can stop them from completing a purchase.
- Overall satisfaction is around **3 out of 5**, so there is room for improvement.

Recommendation:

- Keep prices competitive, reduce cart-abandonment problems, and make the shopping process easier.

Insight: No is the largest response, with 295 customers. Yes has 261 and Sometimes has 244.

Recommendation: Make recommendations more useful and better matched to what customers actually want.



Purchase Frequency

Insight:

- Customers shop at different frequencies, with regular shopping being common.

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Recommendation:

- Give personalized offers and discounts to encourage customers to shop more often.

4. Task 3 – Customer Segmentation and Profiling

```
sns.barplot(x=segment_counts.index, y=segment_counts.values, palette= blues_r )
```



Customers were grouped based on **purchase frequency and shopping satisfaction**.

- **Frequent Buyers:** 128 customers & Avg. Purchase Frequency: **4.48** Avg. Satisfaction: **4.56**
- **Occasional Shoppers:** 352 customers & Avg. Purchase Frequency: **2.45** Avg. Satisfaction: **3.76**
- **At-Risk Customers:** 320 customers & Avg. Purchase Frequency: **3.08** Avg. Satisfaction: **1.51**

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Insight:

- **Frequent Buyers** are the most valuable customers because they buy more often and have high satisfaction.
- **Occasional Shoppers** are the largest group, so they have good potential to become regular buyers.
- **At-Risk Customers** have very low satisfaction, so they are the main group that needs attention.

Recommendation:

- Give **special rewards and offers** to Frequent Buyers to keep them loyal.
- Target Occasional Shoppers with **personalized offers and discounts** to increase their purchase frequency.
- Focus on At-Risk Customers by **finding their problems, improving service, and giving suitable offers** to bring them back.

3. Cart Abandonment

Insight:

- **Better price elsewhere** is a major reason customers leave their carts.

Recommendation:

- Offer competitive prices and simple checkout options to reduce cart abandonment.

4. Rating Accuracy

Insight:

- Most customers give rating accuracy scores around **3/5**.

Recommendation:

- Improve review quality and show more reliable ratings to build customer trust.

Task 4 – Recommendation and Review Insights

- **Recommendation & Satisfaction:** Satisfaction changes only a little, from **2.89 to 3.08**. The correlation is only **0.04**.
- **Review Reliability:** Rating accuracy ranges from **2.69 to 3.29**. Better reviews are linked with better rating accuracy.
- **Personalized Recommendations:** Review importance increases from **2.89 to 3.02**, while rating accuracy slightly decreases from **3.00 to 2.89**.

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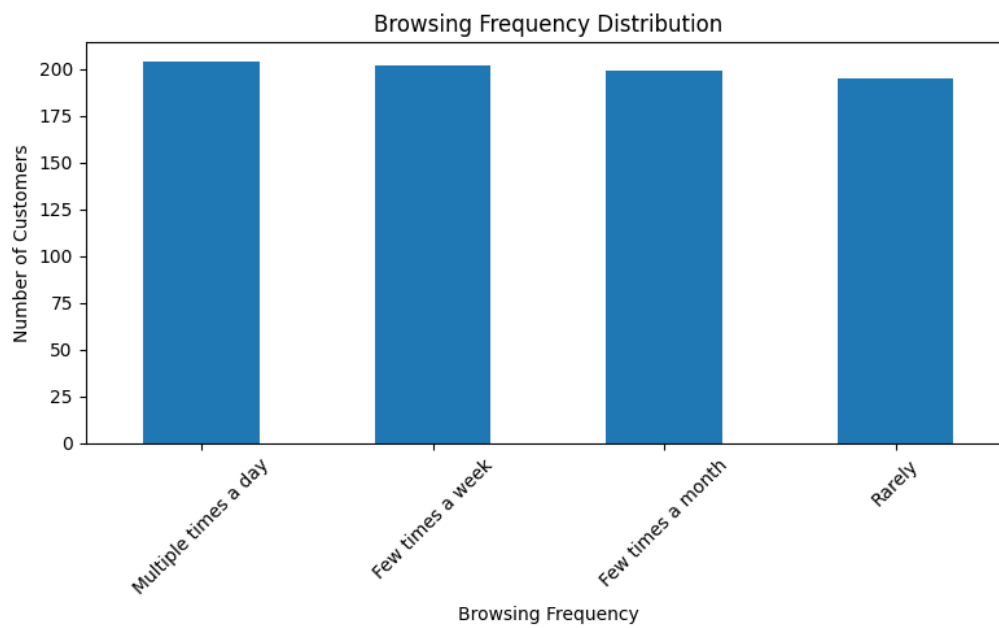
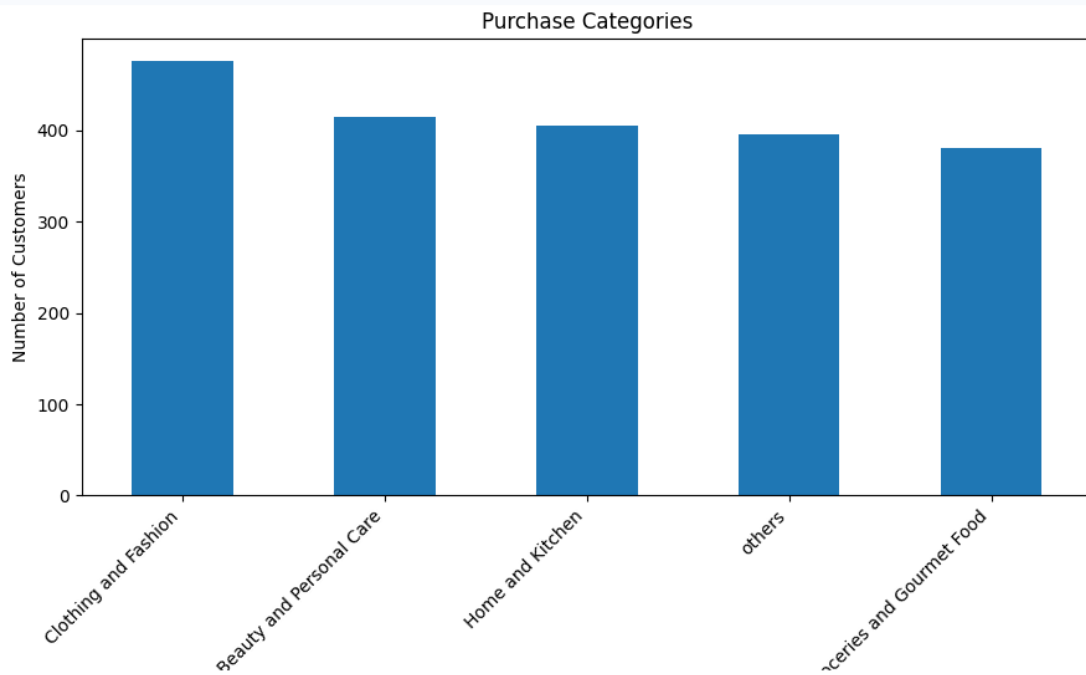
Insight:

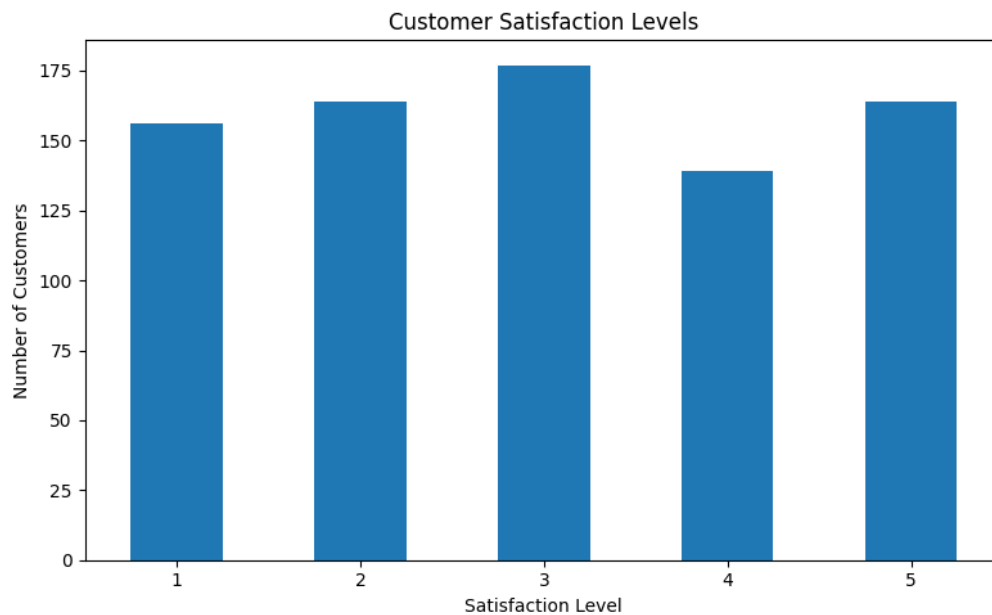
- Recommendations have a **very weak impact on satisfaction**.
- Reliable and helpful reviews improve **rating accuracy and trust**.
- Customers still depend on **reviews and ratings** before buying.

Recommendation:

- Use **past purchases, browsing history, and ratings** for better personalization.
- Show **verified reviews and clear ratings** with recommendations.
- Improve recommendation relevance and **customer trust**.

5.1 Purchase Category





5.2 Recommendation Helpfulness and Satisfaction

Actual output: Satisfaction was 2.89 for score 1, 3.08 for score 2 and 3.02 for score 3. Correlation = 0.0402.

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Insight: Recommendation helpfulness has almost no direct effect on customer satisfaction in this data.

Recommendation: Do not depend only on recommendations. Better prices, product quality, delivery and customer support should also be improved.

5.3 Review Reliability and Helpfulness

Actual output: Rating accuracy ranged from **2.69 to 3.29** across the groups.

Insight: Customers do not see all reviews as equally reliable or helpful. Better reviews can make product ratings more trustworthy.

Recommendation: Highlight verified, useful and detailed reviews so customers can make better buying decisions.

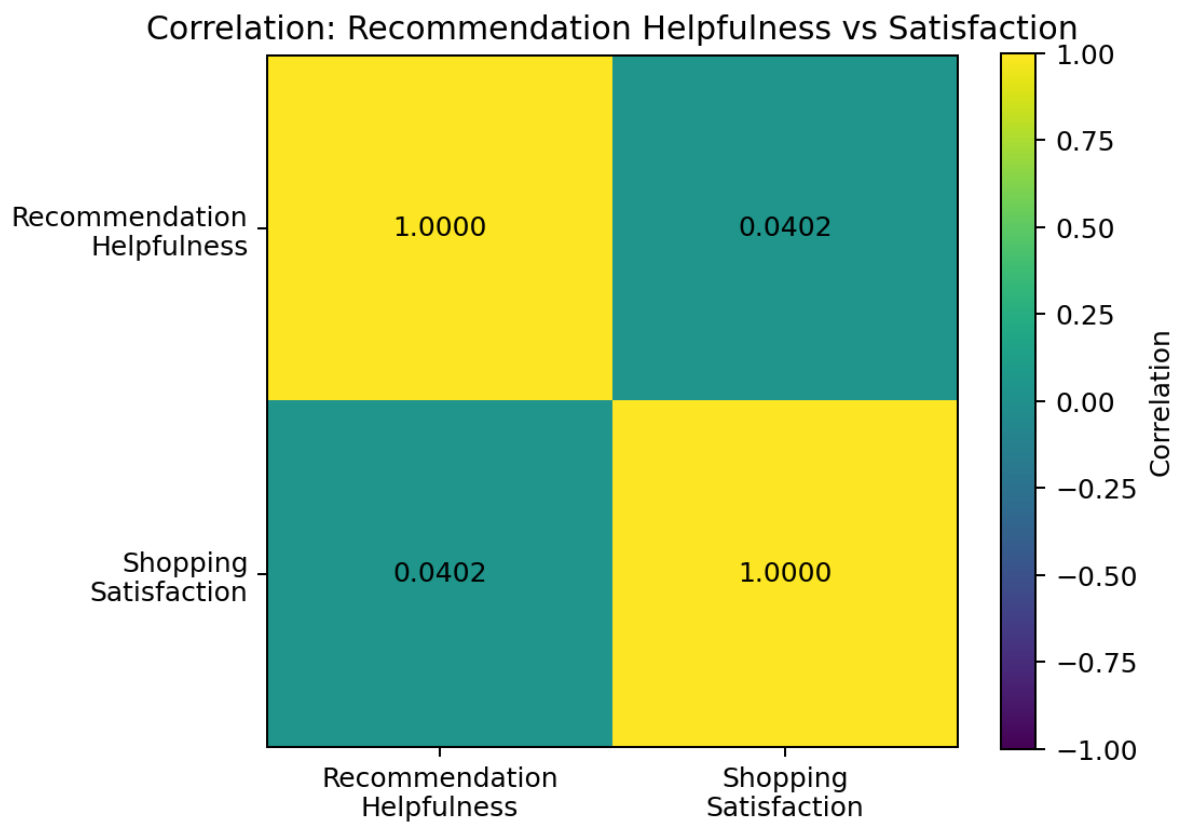
5.4 Personalized Recommendation Use

Actual output:
Did not use = **35.63%**

Sometimes = **34.00%**

Yes = **30.38%**

Sometimes users had the highest helpfulness score of **2.06**.



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Insight: The correlation is 0.0402, which is very weak.

Recommendation: Improve recommendation quality, but also work on price, product quality, delivery and customer service.

Insight: More than one-third of customers do not use personalized recommendations. So, there is a clear chance to improve how these recommendations are shown and matched.

Recommendation: Make personalized suggestions more relevant and easier to notice by using customers' interests, past purchases and ratings.

Business and Financial Risk

The data does not directly measure financial loss. However, it shows some clear business risks. There are 320 At-Risk Customers with average satisfaction of only 1.51. There are also 352 Occasional Shoppers who could become more regular

buyers. Better prices, fewer cart-abandonment problems and a better shopping experience could help protect repeat sales.

Conclusion

This analysis shows that **Snapdeal has many active customers, but there are still some areas that need improvement.** Customers are fairly satisfied, but recommendations are not always helpful, and many customers do not use personalized recommendations regularly.

Snapdeal should focus on **better recommendations, trustworthy reviews, competitive prices, good product quality, smooth delivery, and customer support.** These improvements can help **increase customer satisfaction, build trust, and bring customers back to shop again.**

Thank You